

MSC-SAC-Matis-Sammy

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Chapter 1

Topic Index

1.1 Topics

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Class Index

2.1 Class List

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File Index

3.1 File List

Here is a list of all documented files with brief descriptions:

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Inc/ dma.h	This file contains all the function prototypes for the dma.c file	18
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Inc/ main.h	: Header for main.c file. This file contains the common defines of the application	20
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Chapter 4

Topic Documentation

4.1 CMSIS

Topics

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•	Stm32g4xx_system 7

4.1.1 Detailed Description

4.1.2 Stm32g4xx_system

Topics

•	
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4.1.2.1 Detailed Description

4.1.2.2 STM32G4xx_System_Private_Includes

Macros

- #define [HSE_VALUE](#) 24000000U
- #define [HSI_VALUE](#) 16000000U

4.1.2.2.1 Detailed Description

4.1.2.2.2 Macro Definition Documentation

4.1.2.2.2.1 HSE_VALUE

```
#define HSE_VALUE 24000000U
```

Value of the External oscillator in Hz

4.1.2.2.2.2 HSI_VALUE

```
#define HSI_VALUE 16000000U
```

Value of the Internal oscillator in Hz

4.1.2.3 STM32G4xx_System_Private_TypeDefinitions

4.1.2.4 STM32G4xx_System_Private_Defines

4.1.2.5 STM32G4xx_System_Private_Macros

4.1.2.6 STM32G4xx_System_Private_Variables

Variables

- uint32_t **SystemCoreClock** = [HSI_VALUE](#)
- const uint8_t **AHBPrescTable** [16] = {0U, 0U, 0U, 0U, 0U, 0U, 0U, 0U, 1U, 2U, 3U, 4U, 6U, 7U, 8U, 9U}
- const uint8_t **APBPrescTable** [8] = {0U, 0U, 0U, 0U, 1U, 2U, 3U, 4U}

4.1.2.6.1 Detailed Description

4.1.2.7 STM32G4xx_System_Private_FunctionPrototypes

4.1.2.8 STM32G4xx_System_Private_Functions

Functions

- void [SystemInit](#) (void)
Setup the microcontroller system.
- void [SystemCoreClockUpdate](#) (void)
Update SystemCoreClock variable according to Clock Register Values. The SystemCoreClock variable contains the core clock (HCLK), it can be used by the user application to setup the SysTick timer or configure other parameters.

4.1.2.8.1 Detailed Description

4.1.2.8.2 Function Documentation

4.1.2.8.2.1 SystemCoreClockUpdate()

```
void SystemCoreClockUpdate (  
    void )
```

Update SystemCoreClock variable according to Clock Register Values. The SystemCoreClock variable contains the core clock (HCLK), it can be used by the user application to setup the SysTick timer or configure other parameters.

Note

Each time the core clock (HCLK) changes, this function must be called to update SystemCoreClock variable value. Otherwise, any configuration based on this variable will be incorrect.

- The system frequency computed by this function is not the real frequency in the chip. It is calculated based on the predefined constant and the selected clock source:

- If SYSCLK source is HSI, SystemCoreClock will contain the [HSI_VALUE\(**\)](#)
- If SYSCLK source is HSE, SystemCoreClock will contain the [HSE_VALUE\(***\)](#)
- If SYSCLK source is PLL, SystemCoreClock will contain the [HSE_VALUE\(***\)](#) or [HSI_VALUE\(*\)](#) multiplied/divided by the PLL factors.

(**) HSI_VALUE is a constant defined in stm32g4xx_hal.h file (default value 16 MHz) but the real value may vary depending on the variations in voltage and temperature.

(***) HSE_VALUE is a constant defined in stm32g4xx_hal.h file (default value 24 MHz), user has to ensure that HSE_VALUE is same as the real frequency of the crystal used. Otherwise, this function may have wrong result.

- The result of this function could be not correct when using fractional value for HSE crystal.

Parameters

None	
------	--

Return values

None	
------	--

4.1.2.8.2.2 SystemInit()

```
void SystemInit (
    void )
```

Setup the microcontroller system.

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

Chapter 5

Class Documentation

5.1 drv_shell_struct Struct Reference

Structure representing a shell driver.

```
#include <shell.h>
```

Public Attributes

- [drv_shell_transmit_t](#) transmit
- [drv_shell_receive_t](#) receive

5.1.1 Detailed Description

Structure representing a shell driver.

5.1.2 Member Data Documentation

5.1.2.1 receive

```
drv\_shell\_receive\_t drv_shell_struct::receive
```

Function pointer for receiving data.

5.1.2.2 transmit

```
drv\_shell\_transmit\_t drv_shell_struct::transmit
```

Function pointer for transmitting data.

The documentation for this struct was generated from the following file:

- Inc/user_interface/[shell.h](#)

5.2 h_shell_struct Struct Reference

Structure representing a shell instance.

```
#include <shell.h>
```

Public Attributes

- int [func_list_size](#)
- [shell_func_t](#) [func_list](#) [[SHELL_FUNC_LIST_MAX_SIZE](#)]
- char [print_buffer](#) [[SHELL_PRINT_BUFFER_SIZE](#)]
- char [cmd_buffer](#) [[SHELL_CMD_BUFFER_SIZE](#)]
- [drv_shell_t](#) [drv](#)

5.2.1 Detailed Description

Structure representing a shell instance.

5.2.2 Member Data Documentation

5.2.2.1 cmd_buffer

```
char h_shell_struct::cmd_buffer[SHELL_CMD_BUFFER_SIZE]
```

Buffer used for input.

5.2.2.2 drv

```
drv_shell_t h_shell_struct::drv
```

The shell driver instance.

5.2.2.3 func_list

```
shell_func_t h_shell_struct::func_list[SHELL_FUNC_LIST_MAX_SIZE]
```

Array of registered shell functions.

5.2.2.4 func_list_size

```
int h_shell_struct::func_list_size
```

The current size of the function list.

5.2.2.5 print_buffer

```
char h_shell_struct::print_buffer[SHELL_PRINT_BUFFER_SIZE]
```

Buffer used for output.

The documentation for this struct was generated from the following file:

- Inc/user_interface/[shell.h](#)

5.3 PI_Controller Struct Reference

Structure représentant un correcteur Proportionnel-Intégral (PI).

```
#include <asserv.h>
```

Public Attributes

- float **Kp**
- float **Ki**
- float **limMin**
- float **limMax**
- float **limMinInt**
- float **limMaxInt**
- float **T**
- float **integrator**
- float **prevError**
- float **prevMeasurement**
- float **out**

5.3.1 Detailed Description

Structure représentant un correcteur Proportionnel-Intégral (PI).

Cette structure contient les gains, les limites de saturation (sortie et intégrateur), la période d'échantillonnage ainsi que les variables d'état (mémoire) nécessaires au calcul de la boucle de régulation.

The documentation for this struct was generated from the following file:

- Inc/motor_control/asserv.h

5.4 shell_func_struct Struct Reference

Structure representing a shell command.

```
#include <shell.h>
```

Public Attributes

- char * [string_func_code](#)
- [shell_func_pointer_t](#) func
- char * [description](#)

5.4.1 Detailed Description

Structure representing a shell command.

5.4.2 Member Data Documentation

5.4.2.1 description

```
char* shell_func_struct::description
```

The description of the command.

5.4.2.2 func

```
shell\_func\_pointer\_t shell_func_struct::func
```

Pointer to the function implementing the command.

5.4.2.3 string_func_code

```
char* shell_func_struct::string_func_code
```

The character trigger for the command.

The documentation for this struct was generated from the following file:

- Inc/user_interface/[shell.h](#)

Chapter 6

File Documentation

6.1 input_analog.h

```
00001 /*
00002  * input_analog.h
00003  *
00004  * Created on: Nov 11, 2025
00005  * Author: nicolas
00006  */
00007
00008 #ifndef INC_INPUT_ANALOG_H_
00009 #define INC_INPUT_ANALOG_H_
00010
00011 #include "main.h"
00012 #include "user_interface/shell.h"
00013 #include <stdio.h>
00014 #include <string.h>
00015 #include "app.h"
00016
00017 extern ADC_HandleTypeDef hadcl;
00018
00019 //define pour la mesure du courant
00020 #define ADC_RESOLUTION 4096.0f
00021 #define ADC_VREF 3.3f
00022 #define SENSOR_SENSITIVITY 0.025f // GO-SME 10 (0.025 mV/A)
00023 #define V_OFFSET_THEO (ADC_VREF / 2.0f)
00024
00025 int input_analog_init();
00026 float Calculer_Courant_Moyen();
00027
00028 #endif /* INC_INPUT_ANALOG_H_ */
```

6.2 input_encoder.h

```
00001 /*
00002  * input_encoder.h
00003  *
00004  * Created on: Nov 11, 2025
00005  * Author: nicolas
00006  */
00007
00008 #ifndef INC_ACQUISITION_INPUT_ENCODER_H_
00009 #define INC_ACQUISITION_INPUT_ENCODER_H_
00010
00011 #include <stdint.h>
00012 #include "tim.h"
00013 #include "user_interface/shell.h"
00014 #include "./app.h"
00015
00016
00017 #define SENS_ROTATION_CORRECTION (-1.0f) //l'encodeur est inversé par rapport au moteur
00018 #define USECOND_SECOND_CONVERSION 1000000.0f
00019 #define SECOND_MINUTE_CONVERSION 60.0f
00020 #define STEP_ROTATION_CONVERSION 4096.0f // 4096 pas = 1 tour with this encoder mode
00021 #define PI 3.1415
00022
```

```
00023
00024 extern uint16_t ancien_temps_us;
00025 extern uint16_t ancien_CNT_encodeur;
00026 extern float rpm;
00027
00028 int input_encoder_init();
00029 int encoder_show_speed(h_shell_t* h_shell, int argc, char** argv);
00030 void Calcul_Vitesse_Precise(void);
00031
00032 #endif /* INC_ACQUISITION_INPUT_ENCODER_H_ */
```

6.3 Inc/adc.h File Reference

This file contains all the function prototypes for the [adc.c](#) file.

```
#include "main.h"
```

Functions

- void [MX_ADC1_Init](#) (void)
- void [MX_ADC2_Init](#) (void)

Variables

- ADC_HandleTypeDef **hadc1**
- ADC_HandleTypeDef **hadc2**

6.3.1 Detailed Description

This file contains all the function prototypes for the [adc.c](#) file.

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6.3.2 Function Documentation

6.3.2.1 MX_ADC1_Init()

```
void MX_ADC1_Init (
    void )
```

Common config

Configure the ADC multi-mode

Configure Regular Channel

6.3.2.2 MX_ADC2_Init()

```
void MX_ADC2_Init (
    void )
```

Common config

Configure Regular Channel

6.4 adc.h

[Go to the documentation of this file.](#)

```
00001 /* USER CODE BEGIN Header */
00019 /* USER CODE END Header */
00020 /* Define to prevent recursive inclusion -----*/
00021 #ifndef __ADC_H__
00022 #define __ADC_H__
00023
00024 #ifdef __cplusplus
00025 extern "C" {
00026 #endif
00027
00028 /* Includes -----*/
00029 #include "main.h"
00030
00031 /* USER CODE BEGIN Includes */
00032
00033 /* USER CODE END Includes */
00034
00035 extern ADC_HandleTypeDef hadc1;
00036
00037 extern ADC_HandleTypeDef hadc2;
00038
00039 /* USER CODE BEGIN Private defines */
00040
00041 /* USER CODE END Private defines */
00042
00043 void MX_ADC1_Init(void);
00044 void MX_ADC2_Init(void);
00045
00046 /* USER CODE BEGIN Prototypes */
00047
00048 /* USER CODE END Prototypes */
00049
00050 #ifdef __cplusplus
00051 }
00052 #endif
00053
00054 #endif /* __ADC_H__ */
00055
```

6.5 app.h

```
00001 /*
00002  * app.h
00003  *
00004  * Created on: Nov 11, 2025
00005  * Author: nicolas
00006  */
00007
00008 #ifndef INC_APP_H_
00009 #define INC_APP_H_
00010
00011 #include "stm32g4xx_hal.h"
00012 #include "usart.h"
00013
00014 #include "../motor_control/motor.h"
00015 #include "../user_interface/led.h"
00016 #include "../motor_control/asserv.h"
00017 #include "tim.h"
00018 #include "user_interface/shell.h"
00019 #include "../acquisition/input_analog.h"
```

```

00020
00021
00022
00023 #include "../acquisition/input_encoder.h"
00024 #define VDC 48
00025 #define SPEED_ASSERV_PERIOD 50
00026
00027 extern int start_asserv_flag ;
00028 extern PI_Controller Current_PI_Controller ;
00029 extern int current_asserv ;
00030 extern int speed_asserv ;
00031
00032 void init_device();
00033 void loop();
00034
00035 uint8_t shell_uart2_transmit(const char *pData, uint16_t size);
00036 uint8_t shell_uart2_receive(char *pData, uint16_t size);
00037
00038
00039 #endif /* INC_APP_H_ */

```

6.6 Inc/dma.h File Reference

This file contains all the function prototypes for the [dma.c](#) file.

```
#include "main.h"
```

Functions

- void [MX_DMA_Init](#) (void)

6.6.1 Detailed Description

This file contains all the function prototypes for the [dma.c](#) file.

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6.6.2 Function Documentation

6.6.2.1 MX_DMA_Init()

```
void MX_DMA_Init (
    void )
```

Enable DMA controller clock

6.7 dma.h

[Go to the documentation of this file.](#)

```

00001 /* USER CODE BEGIN Header */
00019 /* USER CODE END Header */
00020 /* Define to prevent recursive inclusion -----*/
00021 #ifndef __DMA_H__
00022 #define __DMA_H__
00023
00024 #ifdef __cplusplus
00025 extern "C" {
00026 #endif
00027
00028 /* Includes -----*/
00029 #include "main.h"
00030
00031 /* DMA memory to memory transfer handles -----*/
00032
00033 /* USER CODE BEGIN Includes */
00034
00035 /* USER CODE END Includes */
00036
00037 /* USER CODE BEGIN Private defines */
00038
00039 /* USER CODE END Private defines */
00040
00041 void MX_DMA_Init(void);
00042
00043 /* USER CODE BEGIN Prototypes */
00044
00045 /* USER CODE END Prototypes */
00046
00047 #ifdef __cplusplus
00048 }
00049 #endif
00050
00051 #endif /* __DMA_H__ */
00052

```

6.8 Inc/gpio.h File Reference

This file contains all the function prototypes for the [gpio.c](#) file.

```
#include "main.h"
```

Functions

- void [MX_GPIO_Init](#) (void)

6.8.1 Detailed Description

This file contains all the function prototypes for the [gpio.c](#) file.

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6.8.2 Function Documentation

6.8.2.1 MX_GPIO_Init()

```
void MX_GPIO_Init (
    void )
```

Configure pins as Analog Input Output EVENT_OUT EXTI

6.9 gpio.h

[Go to the documentation of this file.](#)

```
00001 /* USER CODE BEGIN Header */
00019 /* USER CODE END Header */
00020 /* Define to prevent recursive inclusion -----*/
00021 #ifndef __GPIO_H__
00022 #define __GPIO_H__
00023
00024 #ifdef __cplusplus
00025 extern "C" {
00026 #endif
00027
00028 /* Includes -----*/
00029 #include "main.h"
00030
00031 /* USER CODE BEGIN Includes */
00032
00033 /* USER CODE END Includes */
00034
00035 /* USER CODE BEGIN Private defines */
00036
00037 /* USER CODE END Private defines */
00038
00039 void MX_GPIO_Init(void);
00040
00041 /* USER CODE BEGIN Prototypes */
00042
00043 /* USER CODE END Prototypes */
00044
00045 #ifdef __cplusplus
00046 }
00047 #endif
00048 #endif /*__ GPIO_H__ */
00049
```

6.10 Inc/main.h File Reference

: Header for [main.c](#) file. This file contains the common defines of the application.

```
#include "stm32g4xx_hal.h"
```

Macros

- **#define USR_BTN_Pin** GPIO_PIN_13
- **#define USR_BTN_GPIO_Port** GPIOC
- **#define USR_BTN_EXTI_IRQn** EXTI15_10_IRQn
- **#define U_VPh_Pin** GPIO_PIN_0
- **#define U_VPh_GPIO_Port** GPIOC
- **#define W_VPh_Pin** GPIO_PIN_1
- **#define W_VPh_GPIO_Port** GPIOC

- #define **Bus_Imes_Pin** GPIO_PIN_2
- #define **Bus_Imes_GPIO_Port** GPIOC
- #define **V_VPh_Pin** GPIO_PIN_3
- #define **V_VPh_GPIO_Port** GPIOC
- #define **Bus_V_Pin** GPIO_PIN_0
- #define **Bus_V_GPIO_Port** GPIOA
- #define **U_Imes_Pin** GPIO_PIN_1
- #define **U_Imes_GPIO_Port** GPIOA
- #define **USR_LED_Pin** GPIO_PIN_5
- #define **USR_LED_GPIO_Port** GPIOA
- #define **U_PWM_L_Pin** GPIO_PIN_13
- #define **U_PWM_L_GPIO_Port** GPIOB
- #define **V_PWM_L_Pin** GPIO_PIN_14
- #define **V_PWM_L_GPIO_Port** GPIOB
- #define **W_PWM_L_Pin** GPIO_PIN_15
- #define **W_PWM_L_GPIO_Port** GPIOB
- #define **Enc_Z_Pin** GPIO_PIN_8
- #define **Enc_Z_GPIO_Port** GPIOC
- #define **Enc_Z_EXTI_IRQn** EXTI9_5_IRQn
- #define **U_PWM_H_Pin** GPIO_PIN_8
- #define **U_PWM_H_GPIO_Port** GPIOA
- #define **V_PWM_H_Pin** GPIO_PIN_9
- #define **V_PWM_H_GPIO_Port** GPIOA
- #define **W_PWM_H_Pin** GPIO_PIN_10
- #define **W_PWM_H_GPIO_Port** GPIOA
- #define **T_SWDIO_Pin** GPIO_PIN_13
- #define **T_SWDIO_GPIO_Port** GPIOA
- #define **T_SWCLK_Pin** GPIO_PIN_14
- #define **T_SWCLK_GPIO_Port** GPIOA
- #define **NRST_SafetyUC_Pin** GPIO_PIN_2
- #define **NRST_SafetyUC_GPIO_Port** GPIOD
- #define **T_SWO_Pin** GPIO_PIN_3
- #define **T_SWO_GPIO_Port** GPIOB

Functions

- void [Error_Handler](#) (void)
This function is executed in case of error occurrence.

6.10.1 Detailed Description

: Header for [main.c](#) file. This file contains the common defines of the application.

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6.10.2 Function Documentation

6.10.2.1 Error_Handler()

```
void Error_Handler (
    void )
```

This function is executed in case of error occurrence.

Return values

None	
------	--

6.11 main.h

[Go to the documentation of this file.](#)

```
00001 /* USER CODE BEGIN Header */
00019 /* USER CODE END Header */
00020
00021 /* Define to prevent recursive inclusion -----*/
00022 #ifndef __MAIN_H
00023 #define __MAIN_H
00024
00025 #ifdef __cplusplus
00026 extern "C" {
00027 #endif
00028
00029 /* Includes -----*/
00030 #include "stm32g4xx_hal.h"
00031
00032 /* Private includes -----*/
00033 /* USER CODE BEGIN Includes */
00034
00035 /* USER CODE END Includes */
00036
00037 /* Exported types -----*/
00038 /* USER CODE BEGIN ET */
00039
00040 /* USER CODE END ET */
00041
00042 /* Exported constants -----*/
00043 /* USER CODE BEGIN EC */
00044
00045 /* USER CODE END EC */
00046
00047 /* Exported macro -----*/
00048 /* USER CODE BEGIN EM */
00049
00050 /* USER CODE END EM */
00051
00052 /* Exported functions prototypes -----*/
00053 void Error_Handler(void);
00054
00055 /* USER CODE BEGIN EFP */
00056
00057 /* USER CODE END EFP */
00058
00059 /* Private defines -----*/
00060 #define USR_BTN_Pin GPIO_PIN_13
00061 #define USR_BTN_GPIO_Port GPIOC
00062 #define USR_BTN_EXTI_IRQn EXTI15_10_IRQn
00063 #define U_VPh_Pin GPIO_PIN_0
00064 #define U_VPh_GPIO_Port GPIOC
00065 #define W_VPh_Pin GPIO_PIN_1
00066 #define W_VPh_GPIO_Port GPIOC
00067 #define Bus_Imes_Pin GPIO_PIN_2
00068 #define Bus_Imes_GPIO_Port GPIOC
00069 #define V_VPh_Pin GPIO_PIN_3
00070 #define V_VPh_GPIO_Port GPIOC
00071 #define Bus_V_Pin GPIO_PIN_0
00072 #define Bus_V_GPIO_Port GPIOA
00073 #define U_Imes_Pin GPIO_PIN_1
00074 #define U_Imes_GPIO_Port GPIOA
```

```

00075 #define USR_LED_Pin GPIO_PIN_5
00076 #define USR_LED_GPIO_Port GPIOA
00077 #define U_PWM_L_Pin GPIO_PIN_13
00078 #define U_PWM_L_GPIO_Port GPIOB
00079 #define V_PWM_L_Pin GPIO_PIN_14
00080 #define V_PWM_L_GPIO_Port GPIOB
00081 #define W_PWM_L_Pin GPIO_PIN_15
00082 #define W_PWM_L_GPIO_Port GPIOB
00083 #define Enc_Z_Pin GPIO_PIN_8
00084 #define Enc_Z_GPIO_Port GPIOC
00085 #define Enc_Z_EXTI_IRQn EXTI9_5_IRQn
00086 #define U_PWM_H_Pin GPIO_PIN_8
00087 #define U_PWM_H_GPIO_Port GPIOA
00088 #define V_PWM_H_Pin GPIO_PIN_9
00089 #define V_PWM_H_GPIO_Port GPIOA
00090 #define W_PWM_H_Pin GPIO_PIN_10
00091 #define W_PWM_H_GPIO_Port GPIOA
00092 #define T_SWDIO_Pin GPIO_PIN_13
00093 #define T_SWDIO_GPIO_Port GPIOA
00094 #define T_SWCLK_Pin GPIO_PIN_14
00095 #define T_SWCLK_GPIO_Port GPIOA
00096 #define NRST_SafetyUC_Pin GPIO_PIN_2
00097 #define NRST_SafetyUC_GPIO_Port GPIOB
00098 #define T_SWO_Pin GPIO_PIN_3
00099 #define T_SWO_GPIO_Port GPIOB
00100
00101 /* USER CODE BEGIN Private defines */
00102 /* USER CODE END Private defines */
00103
00104 #ifdef __cplusplus
00105 }
00106 #endif
00107
00108 #endif /* __MAIN_H */

```

6.12 asserv.h

```

00001 /*
00002  * asserv.h
00003  *
00004  * Created on: Nov 11, 2025
00005  * Author: nicolas
00006  */
00007
00008 #ifndef INC_MOTOR_CONTROL_ASSERV_H_
00009 #define INC_MOTOR_CONTROL_ASSERV_H_
00010
00011
00012 typedef struct {
00013
00014     /* Controller gains */
00015     float Kp;
00016     float Ki;
00017
00018     /* Output limits */
00019     float limMin;
00020     float limMax;
00021
00022     /* Integrator limits */
00023     float limMinInt;
00024     float limMaxInt;
00025
00026     /* Sample time (in seconds) */
00027     float T;
00028
00029     /* Controller "memory" */
00030     float integrator;
00031     float prevError; /* Required for integrator */
00032     float prevMeasurement; /* Required for differentiator */
00033
00034     /* Controller output */
00035     float out;
00036 } PI_Controller;
00037
00038 #define VDC 48 //doublon
00039
00040 void Current_PI_Controller_Init(PI_Controller *pi);
00041 void Speed_PI_Controller_Init(PI_Controller *pi);
00042 float PI_Controller_Update(PI_Controller *pi, float setpoint, float measurement);
00043
00044 #endif /* INC_MOTOR_CONTROL_ASSERV_H_ */

```

6.13 motor.h

```

00001 /*
00002  * motor.h
00003  *
00004  * Created on: Nov 11, 2025
00005  * Author: nicolas
00006  */
00007
00008 #ifndef INC_MOTOR_CONTROL_MOTOR_H_
00009 #define INC_MOTOR_CONTROL_MOTOR_H_
00010
00011 #define ARR_VALUE 8499
00012 #define PSC_VALUE 0
00013 #define COMMAND_MAX_VALUE 8500
00014 #define COMMAND_HALF_MAX_VALUE 4250
00015 #define TWO_ARGUMENTS 2
00016 #define DELAY_RAMP_CRR_VALUE 50000 // less than 1 us
00017 #define INCREMENT 200
00018 #define PERCENT_TO_MAX_CRR_VALUE_CONVERSION 85 // conversion to max CRR Value
00019 #define DUTY_CYCLE_MAX_VALUE 100
00020 #define MOTOR_SPEED_MAX_VALUE 3000
00021 #define PI 3.1415
00022
00023 #include <stdint.h>
00024 #include "tim.h"
00025 #include "user_interface/shell.h"
00026 #include "./app.h"
00027
00028 extern int motor_command;
00029
00030 int motor_init();
00031
00032 #endif /* INC_MOTOR_CONTROL_MOTOR_H_ */

```

6.14 Inc/stm32g4xx_hal_conf.h File Reference

HAL configuration file.

```

#include "stm32g4xx_hal_rcc.h"
#include "stm32g4xx_hal_gpio.h"
#include "stm32g4xx_hal_dma.h"
#include "stm32g4xx_hal_cortex.h"
#include "stm32g4xx_hal_adc.h"
#include "stm32g4xx_hal_exti.h"
#include "stm32g4xx_hal_flash.h"
#include "stm32g4xx_hal_pwr.h"
#include "stm32g4xx_hal_tim.h"
#include "stm32g4xx_hal_uart.h"

```

Macros

- **#define HAL_MODULE_ENABLED**
This is the list of modules to be used in the HAL driver.
- **#define HAL_ADC_MODULE_ENABLED**
- **#define HAL_TIM_MODULE_ENABLED**
- **#define HAL_UART_MODULE_ENABLED**
- **#define HAL_GPIO_MODULE_ENABLED**
- **#define HAL_EXTI_MODULE_ENABLED**
- **#define HAL_DMA_MODULE_ENABLED**
- **#define HAL_RCC_MODULE_ENABLED**
- **#define HAL_FLASH_MODULE_ENABLED**
- **#define HAL_PWR_MODULE_ENABLED**

- `#define HAL_CORTEX_MODULE_ENABLED`
- `#define USE_HAL_ADC_REGISTER_CALLBACKS 0U`
This is the list of modules where register callback can be used.
- `#define USE_HAL_COMP_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_CORDIC_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_CRYPT_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_DAC_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_EXTI_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_FDCAN_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_FMAC_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_HRTIM_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_I2C_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_I2S_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_IRDA_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_LPTIM_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_NAND_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_NOR_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_OPAMP_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_PCD_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_QSPI_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_RNG_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_RTC_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_SAI_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_SMARTCARD_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_SMBUS_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_SPI_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_SRAM_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_TIM_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_UART_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_USART_REGISTER_CALLBACKS 0U`
- `#define USE_HAL_WWDG_REGISTER_CALLBACKS 0U`
- `#define HSE_VALUE (24000000UL)`
Adjust the value of External High Speed oscillator (HSE) used in your application. This value is used by the RCC HAL module to compute the system frequency (when HSE is used as system clock source, directly or through the PLL).
- `#define HSE_STARTUP_TIMEOUT (100UL)`
- `#define HSI_VALUE (16000000UL)`
Internal High Speed oscillator (HSI) value. This value is used by the RCC HAL module to compute the system frequency (when HSI is used as system clock source, directly or through the PLL).
- `#define HSI48_VALUE (48000000UL)`
Internal High Speed oscillator (HSI48) value for USB FS and RNG. This internal oscillator is mainly dedicated to provide a high precision clock to the USB peripheral by means of a special Clock Recovery System (CRS) circuitry. When the CRS is not used, the HSI48 RC oscillator runs on it default frequency which is subject to manufacturing process variations.
- `#define LSI_VALUE (32000UL)`
Internal Low Speed oscillator (LSI) value.
- `#define LSE_VALUE (32768UL)`
External Low Speed oscillator (LSE) value. This value is used by the UART, RTC HAL module to compute the system frequency.
- `#define LSE_STARTUP_TIMEOUT (5000UL)`
- `#define EXTERNAL_CLOCK_VALUE (12288000UL)`
External clock source for I2S and SAI peripherals This value is used by the I2S and SAI HAL modules to compute the I2S and SAI clock source frequency, this source is inserted directly through I2S_CKIN pad.
- `#define VDD_VALUE (3300UL)`
This is the HAL system configuration section.

- `#define TICK_INT_PRIORITY (15UL)`
- `#define USE_RTOS 0U`
- `#define PREFETCH_ENABLE 0U`
- `#define INSTRUCTION_CACHE_ENABLE 1U`
- `#define DATA_CACHE_ENABLE 1U`
- `#define USE_SPI_CRC 0U`
Uncomment the line below to expanse the "assert_param" macro in the HAL drivers code.
- `#define assert_param(expr)`
Include module's header file.

6.14.1 Detailed Description

HAL configuration file.

Author

MCD Application Team

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6.14.2 Macro Definition Documentation

6.14.2.1 assert_param

```
#define assert_param(  
    expr)
```

Value:

((void) 0U)

Include module's header file.

6.14.2.2 EXTERNAL_CLOCK_VALUE

```
#define EXTERNAL_CLOCK_VALUE (12288000UL)
```

External clock source for I2S and SAI peripherals This value is used by the I2S and SAI HAL modules to compute the I2S and SAI clock source frequency, this source is inserted directly through I2S_CKIN pad.

Value of the External oscillator in Hz

6.14.2.3 HSE_STARTUP_TIMEOUT

```
#define HSE_STARTUP_TIMEOUT (100UL)
```

Time out for HSE start up, in ms

6.14.2.4 HSE_VALUE

```
#define HSE_VALUE (24000000UL)
```

Adjust the value of External High Speed oscillator (HSE) used in your application. This value is used by the RCC HAL module to compute the system frequency (when HSE is used as system clock source, directly or through the PLL).

Value of the External oscillator in Hz

6.14.2.5 HSI48_VALUE

```
#define HSI48_VALUE (48000000UL)
```

Internal High Speed oscillator (HSI48) value for USB FS and RNG. This internal oscillator is mainly dedicated to provide a high precision clock to the USB peripheral by means of a special Clock Recovery System (CRS) circuitry. When the CRS is not used, the HSI48 RC oscillator runs on it default frequency which is subject to manufacturing process variations.

Value of the Internal High Speed oscillator for USB FS/RNG in Hz. The real value may vary depending on manufacturing process variations.

6.14.2.6 HSI_VALUE

```
#define HSI_VALUE (16000000UL)
```

Internal High Speed oscillator (HSI) value. This value is used by the RCC HAL module to compute the system frequency (when HSI is used as system clock source, directly or through the PLL).

Value of the Internal oscillator in Hz

6.14.2.7 LSE_STARTUP_TIMEOUT

```
#define LSE_STARTUP_TIMEOUT (5000UL)
```

Time out for LSE start up, in ms

6.14.2.8 LSE_VALUE

```
#define LSE_VALUE (32768UL)
```

External Low Speed oscillator (LSE) value. This value is used by the UART, RTC HAL module to compute the system frequency.

Value of the External Low Speed oscillator in Hz

6.14.2.9 LSI_VALUE

```
#define LSI_VALUE (32000UL)
```

Internal Low Speed oscillator (LSI) value.

< Value of the Internal Low Speed oscillator in Hz The real value may vary depending on the variations in voltage and temperature. LSI Typical Value in Hz

6.14.2.10 TICK_INT_PRIORITY

```
#define TICK_INT_PRIORITY (15UL)
```

tick interrupt priority (lowest by default)

6.14.2.11 VDD_VALUE

```
#define VDD_VALUE (3300UL)
```

This is the HAL system configuration section.

Value of VDD in mv

6.15 stm32g4xx_hal_conf.h

[Go to the documentation of this file.](#)

```
00001 /* USER CODE BEGIN Header */
00019 /* USER CODE END Header */
00020
00021 /* Define to prevent recursive inclusion -----*/
00022 #ifndef STM32G4xx_HAL_CONF_H
00023 #define STM32G4xx_HAL_CONF_H
00024
00025 #ifdef __cplusplus
00026 extern "C" {
00027 #endif
00028
00029 /* Exported types -----*/
00030 /* Exported constants -----*/
00031
00032 /* ##### Module Selection ##### */
00036
00037 #define HAL_MODULE_ENABLED
00038
00039 #define HAL_ADC_MODULE_ENABLED
00040 /**\*define HAL_COMP_MODULE_ENABLED */
00041 /**\*define HAL_CORDIC_MODULE_ENABLED */
00042 /**\*define HAL_CRC_MODULE_ENABLED */
00043 /**\*define HAL_Cryp_MODULE_ENABLED */
00044 /**\*define HAL_DAC_MODULE_ENABLED */
00045 /**\*define HAL_FDCAN_MODULE_ENABLED */
00046 /**\*define HAL_FMAC_MODULE_ENABLED */
00047 /**\*define HAL_HRTIM_MODULE_ENABLED */
00048 /**\*define HAL_IRDA_MODULE_ENABLED */
00049 /**\*define HAL_IWDG_MODULE_ENABLED */
00050 /**\*define HAL_I2C_MODULE_ENABLED */
00051 /**\*define HAL_I2S_MODULE_ENABLED */
00052 /**\*define HAL_LPTIM_MODULE_ENABLED */
00053 /**\*define HAL_NAND_MODULE_ENABLED */
00054 /**\*define HAL_NOR_MODULE_ENABLED */
00055 /**\*define HAL_OPAMP_MODULE_ENABLED */
00056 /**\*define HAL_PCD_MODULE_ENABLED */
00057 /**\*define HAL_QSPI_MODULE_ENABLED */
00058 /**\*define HAL_RNG_MODULE_ENABLED */
```

```

00059 /**define HAL_RTC_MODULE_ENABLED */
00060 /**define HAL_SAI_MODULE_ENABLED */
00061 /**define HAL_SMARTCARD_MODULE_ENABLED */
00062 /**define HAL_SMBUS_MODULE_ENABLED */
00063 /**define HAL_SPI_MODULE_ENABLED */
00064 /**define HAL_SRAM_MODULE_ENABLED */
00065 #define HAL_TIM_MODULE_ENABLED
00066 #define HAL_UART_MODULE_ENABLED
00067 /**define HAL_USART_MODULE_ENABLED */
00068 /**define HAL_WWDG_MODULE_ENABLED */
00069 #define HAL_GPIO_MODULE_ENABLED
00070 #define HAL_EXTI_MODULE_ENABLED
00071 #define HAL_DMA_MODULE_ENABLED
00072 #define HAL_RCC_MODULE_ENABLED
00073 #define HAL_FLASH_MODULE_ENABLED
00074 #define HAL_PWR_MODULE_ENABLED
00075 #define HAL_CORTEX_MODULE_ENABLED
00076
00077 /* ##### Register Callbacks selection ##### */
00081 #define USE_HAL_ADC_REGISTER_CALLBACKS 0U
00082 #define USE_HAL_COMP_REGISTER_CALLBACKS 0U
00083 #define USE_HAL_CORDIC_REGISTER_CALLBACKS 0U
00084 #define USE_HAL_Cryp_REGISTER_CALLBACKS 0U
00085 #define USE_HAL_DAC_REGISTER_CALLBACKS 0U
00086 #define USE_HAL_EXTI_REGISTER_CALLBACKS 0U
00087 #define USE_HAL_FDCAN_REGISTER_CALLBACKS 0U
00088 #define USE_HAL_FMAC_REGISTER_CALLBACKS 0U
00089 #define USE_HAL_HRTIM_REGISTER_CALLBACKS 0U
00090 #define USE_HAL_I2C_REGISTER_CALLBACKS 0U
00091 #define USE_HAL_I2S_REGISTER_CALLBACKS 0U
00092 #define USE_HAL_IRDA_REGISTER_CALLBACKS 0U
00093 #define USE_HAL_LPTIM_REGISTER_CALLBACKS 0U
00094 #define USE_HAL_NAND_REGISTER_CALLBACKS 0U
00095 #define USE_HAL_NOR_REGISTER_CALLBACKS 0U
00096 #define USE_HAL_OPAMP_REGISTER_CALLBACKS 0U
00097 #define USE_HAL_PCD_REGISTER_CALLBACKS 0U
00098 #define USE_HAL_QSPI_REGISTER_CALLBACKS 0U
00099 #define USE_HAL_RNG_REGISTER_CALLBACKS 0U
00100 #define USE_HAL_RTC_REGISTER_CALLBACKS 0U
00101 #define USE_HAL_SAI_REGISTER_CALLBACKS 0U
00102 #define USE_HAL_SMARTCARD_REGISTER_CALLBACKS 0U
00103 #define USE_HAL_SMBUS_REGISTER_CALLBACKS 0U
00104 #define USE_HAL_SPI_REGISTER_CALLBACKS 0U
00105 #define USE_HAL_SRAM_REGISTER_CALLBACKS 0U
00106 #define USE_HAL_TIM_REGISTER_CALLBACKS 0U
00107 #define USE_HAL_UART_REGISTER_CALLBACKS 0U
00108 #define USE_HAL_USART_REGISTER_CALLBACKS 0U
00109 #define USE_HAL_WWDG_REGISTER_CALLBACKS 0U
00110
00111 /* ##### Oscillator Values adaptation #####*/
00117 #if !defined (HSE_VALUE)
00118 #define HSE_VALUE (24000000UL)
00119 #endif /* HSE_VALUE */
00120
00121 #if !defined (HSE_STARTUP_TIMEOUT)
00122 #define HSE_STARTUP_TIMEOUT (100UL)
00123 #endif /* HSE_STARTUP_TIMEOUT */
00124
00130 #if !defined (HSI_VALUE)
00131 #define HSI_VALUE (16000000UL)
00132 #endif /* HSI_VALUE */
00133
00141 #if !defined (HSI48_VALUE)
00142 #define HSI48_VALUE (48000000UL)
00144 #endif /* HSI48_VALUE */
00145
00149 #if !defined (LSI_VALUE)
00152 #define LSI_VALUE (32000UL)
00153 #endif /* LSI_VALUE */
00158 #if !defined (LSE_VALUE)
00159 #define LSE_VALUE (32768UL)
00160 #endif /* LSE_VALUE */
00161
00162 #if !defined (LSE_STARTUP_TIMEOUT)
00163 #define LSE_STARTUP_TIMEOUT (5000UL)
00164 #endif /* LSE_STARTUP_TIMEOUT */
00165
00171 #if !defined (EXTERNAL_CLOCK_VALUE)
00172 #define EXTERNAL_CLOCK_VALUE (12288000UL)
00173 #endif /* EXTERNAL_CLOCK_VALUE */
00174
00175 /* Tip: To avoid modifying this file each time you need to use different HSE,
00176 == you can define the HSE value in your toolchain compiler preprocessor. */
00177
00178 /* ##### System Configuration ##### */
00182
00183 #define VDD_VALUE (3300UL)

```

```

00184 #define TICK_INT_PRIORITY          (15UL)
00185 #define USE_RTOS                     0U
00186 #define PREFETCH_ENABLE              0U
00187 #define INSTRUCTION_CACHE_ENABLE     1U
00188 #define DATA_CACHE_ENABLE           1U
00189
00190 /* ##### Assert Selection ##### */
00195 /* #define USE_FULL_ASSERT           1U */
00196
00197 /* ##### SPI peripheral configuration ##### */
00198
00199 /* CRC FEATURE: Use to activate CRC feature inside HAL SPI Driver
00200  * Activated: CRC code is present inside driver
00201  * Deactivated: CRC code cleaned from driver
00202  */
00203
00204 #define USE_SPI_CRC                   0U
00205
00206 /* Includes -----*/
00210
00211 #ifndef HAL_RCC_MODULE_ENABLED
00212 #include "stm32g4xx_hal_rcc.h"
00213 #endif /* HAL_RCC_MODULE_ENABLED */
00214
00215 #ifndef HAL_GPIO_MODULE_ENABLED
00216 #include "stm32g4xx_hal_gpio.h"
00217 #endif /* HAL_GPIO_MODULE_ENABLED */
00218
00219 #ifndef HAL_DMA_MODULE_ENABLED
00220 #include "stm32g4xx_hal_dma.h"
00221 #endif /* HAL_DMA_MODULE_ENABLED */
00222
00223 #ifndef HAL_CORTEX_MODULE_ENABLED
00224 #include "stm32g4xx_hal_cortex.h"
00225 #endif /* HAL_CORTEX_MODULE_ENABLED */
00226
00227 #ifndef HAL_ADC_MODULE_ENABLED
00228 #include "stm32g4xx_hal_adc.h"
00229 #endif /* HAL_ADC_MODULE_ENABLED */
00230
00231 #ifndef HAL_COMP_MODULE_ENABLED
00232 #include "stm32g4xx_hal_comp.h"
00233 #endif /* HAL_COMP_MODULE_ENABLED */
00234
00235 #ifndef HAL_CORDIC_MODULE_ENABLED
00236 #include "stm32g4xx_hal_cordic.h"
00237 #endif /* HAL_CORDIC_MODULE_ENABLED */
00238
00239 #ifndef HAL_CRC_MODULE_ENABLED
00240 #include "stm32g4xx_hal_crc.h"
00241 #endif /* HAL_CRC_MODULE_ENABLED */
00242
00243 #ifndef HAL_Cryp_MODULE_ENABLED
00244 #include "stm32g4xx_hal_cryp.h"
00245 #endif /* HAL_Cryp_MODULE_ENABLED */
00246
00247 #ifndef HAL_DAC_MODULE_ENABLED
00248 #include "stm32g4xx_hal_dac.h"
00249 #endif /* HAL_DAC_MODULE_ENABLED */
00250
00251 #ifndef HAL_EXTI_MODULE_ENABLED
00252 #include "stm32g4xx_hal_exti.h"
00253 #endif /* HAL_EXTI_MODULE_ENABLED */
00254
00255 #ifndef HAL_FDCAN_MODULE_ENABLED
00256 #include "stm32g4xx_hal_fdcan.h"
00257 #endif /* HAL_FDCAN_MODULE_ENABLED */
00258
00259 #ifndef HAL_FLASH_MODULE_ENABLED
00260 #include "stm32g4xx_hal_flash.h"
00261 #endif /* HAL_FLASH_MODULE_ENABLED */
00262
00263 #ifndef HAL_FMAC_MODULE_ENABLED
00264 #include "stm32g4xx_hal_fmac.h"
00265 #endif /* HAL_FMAC_MODULE_ENABLED */
00266
00267 #ifndef HAL_HRTIM_MODULE_ENABLED
00268 #include "stm32g4xx_hal_hrtim.h"
00269 #endif /* HAL_HRTIM_MODULE_ENABLED */
00270
00271 #ifndef HAL_IRDA_MODULE_ENABLED
00272 #include "stm32g4xx_hal_irda.h"
00273 #endif /* HAL_IRDA_MODULE_ENABLED */
00274
00275 #ifndef HAL_IWDG_MODULE_ENABLED
00276 #include "stm32g4xx_hal_iwdg.h"
00277 #endif /* HAL_IWDG_MODULE_ENABLED */

```

```

00278
00279 #ifndef HAL_I2C_MODULE_ENABLED
00280 #include "stm32g4xx_hal_i2c.h"
00281 #endif /* HAL_I2C_MODULE_ENABLED */
00282
00283 #ifndef HAL_I2S_MODULE_ENABLED
00284 #include "stm32g4xx_hal_i2s.h"
00285 #endif /* HAL_I2S_MODULE_ENABLED */
00286
00287 #ifndef HAL_LPTIM_MODULE_ENABLED
00288 #include "stm32g4xx_hal_lptim.h"
00289 #endif /* HAL_LPTIM_MODULE_ENABLED */
00290
00291 #ifndef HAL_NAND_MODULE_ENABLED
00292 #include "stm32g4xx_hal_nand.h"
00293 #endif /* HAL_NAND_MODULE_ENABLED */
00294
00295 #ifndef HAL_NOR_MODULE_ENABLED
00296 #include "stm32g4xx_hal_nor.h"
00297 #endif /* HAL_NOR_MODULE_ENABLED */
00298
00299 #ifndef HAL_OPAMP_MODULE_ENABLED
00300 #include "stm32g4xx_hal_opamp.h"
00301 #endif /* HAL_OPAMP_MODULE_ENABLED */
00302
00303 #ifndef HAL_PCD_MODULE_ENABLED
00304 #include "stm32g4xx_hal_pcd.h"
00305 #endif /* HAL_PCD_MODULE_ENABLED */
00306
00307 #ifndef HAL_PWR_MODULE_ENABLED
00308 #include "stm32g4xx_hal_pwr.h"
00309 #endif /* HAL_PWR_MODULE_ENABLED */
00310
00311 #ifndef HAL_QSPI_MODULE_ENABLED
00312 #include "stm32g4xx_hal_qspi.h"
00313 #endif /* HAL_QSPI_MODULE_ENABLED */
00314
00315 #ifndef HAL_RNG_MODULE_ENABLED
00316 #include "stm32g4xx_hal_rng.h"
00317 #endif /* HAL_RNG_MODULE_ENABLED */
00318
00319 #ifndef HAL_RTC_MODULE_ENABLED
00320 #include "stm32g4xx_hal_rtc.h"
00321 #endif /* HAL_RTC_MODULE_ENABLED */
00322
00323 #ifndef HAL_SAI_MODULE_ENABLED
00324 #include "stm32g4xx_hal_sai.h"
00325 #endif /* HAL_SAI_MODULE_ENABLED */
00326
00327 #ifndef HAL_SMARTCARD_MODULE_ENABLED
00328 #include "stm32g4xx_hal_smartcard.h"
00329 #endif /* HAL_SMARTCARD_MODULE_ENABLED */
00330
00331 #ifndef HAL_SMBUS_MODULE_ENABLED
00332 #include "stm32g4xx_hal_smbus.h"
00333 #endif /* HAL_SMBUS_MODULE_ENABLED */
00334
00335 #ifndef HAL_SPI_MODULE_ENABLED
00336 #include "stm32g4xx_hal_spi.h"
00337 #endif /* HAL_SPI_MODULE_ENABLED */
00338
00339 #ifndef HAL_SRAM_MODULE_ENABLED
00340 #include "stm32g4xx_hal_sram.h"
00341 #endif /* HAL_SRAM_MODULE_ENABLED */
00342
00343 #ifndef HAL_TIM_MODULE_ENABLED
00344 #include "stm32g4xx_hal_tim.h"
00345 #endif /* HAL_TIM_MODULE_ENABLED */
00346
00347 #ifndef HAL_UART_MODULE_ENABLED
00348 #include "stm32g4xx_hal_uart.h"
00349 #endif /* HAL_UART_MODULE_ENABLED */
00350
00351 #ifndef HAL_USART_MODULE_ENABLED
00352 #include "stm32g4xx_hal_usart.h"
00353 #endif /* HAL_USART_MODULE_ENABLED */
00354
00355 #ifndef HAL_WWDG_MODULE_ENABLED
00356 #include "stm32g4xx_hal_wwdg.h"
00357 #endif /* HAL_WWDG_MODULE_ENABLED */
00358
00359 /* Exported macro -----*/
00360 #ifndef USE_FULL_ASSERT
00361 #define assert_param(expr) ((expr) ? (void)0U : assert_failed((uint8_t *)__FILE__, __LINE__))
00362 #endif /* USE_FULL_ASSERT */
00363 /* Exported functions ----- */
00364 void assert_failed(uint8_t *file, uint32_t line);
00365 #else

```

```

00373 #define assert_param(expr) ((void)0U)
00374 #endif /* USE_FULL_ASSERT */
00375
00376 #ifdef __cplusplus
00377 }
00378 #endif
00379
00380 #endif /* STM32G4xx_HAL_CONF_H */

```

6.16 Inc/stm32g4xx_it.h File Reference

This file contains the headers of the interrupt handlers.

Functions

- void **NMI_Handler** (void)
This function handles Non maskable interrupt.
- void **HardFault_Handler** (void)
This function handles Hard fault interrupt.
- void **MemManage_Handler** (void)
This function handles Memory management fault.
- void **BusFault_Handler** (void)
This function handles Prefetch fault, memory access fault.
- void **UsageFault_Handler** (void)
This function handles Undefined instruction or illegal state.
- void **SVC_Handler** (void)
This function handles System service call via SWI instruction.
- void **DebugMon_Handler** (void)
This function handles Debug monitor.
- void **PendSV_Handler** (void)
This function handles Pendable request for system service.
- void **SysTick_Handler** (void)
This function handles System tick timer.
- void **DMA1_Channel2_IRQHandler** (void)
This function handles DMA1 channel2 global interrupt.
- void **EXTI9_5_IRQHandler** (void)
This function handles EXTI line[9:5] interrupts.
- void **TIM1_UP_TIM16_IRQHandler** (void)
This function handles TIM1 update interrupt and TIM16 global interrupt.
- void **DMA1_Channel1_IRQHandler** (void)
- void **TIM1_TRG_COM_TIM17_IRQHandler** (void)
This function handles TIM1 trigger and commutation interrupts and TIM17 global interrupt.
- void **USART2_IRQHandler** (void)
This function handles USART2 global interrupt / USART2 wake-up interrupt through EXTI line 26.
- void **EXTI15_10_IRQHandler** (void)
This function handles EXTI line[15:10] interrupts.
- void **TIM6_DAC_IRQHandler** (void)
This function handles TIM6 global interrupt, DAC1 and DAC3 channel underrun error interrupts.
- void **TIM7_DAC_IRQHandler** (void)
This function handles TIM7 global interrupt, DAC2 and DAC4 channel underrun error interrupts.

6.16.1 Detailed Description

This file contains the headers of the interrupt handlers.

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6.17 stm32g4xx_it.h

[Go to the documentation of this file.](#)

```

00001 /* USER CODE BEGIN Header */
00018 /* USER CODE END Header */
00019
00020 /* Define to prevent recursive inclusion -----*/
00021 #ifndef __STM32G4xx_IT_H
00022 #define __STM32G4xx_IT_H
00023
00024 #ifdef __cplusplus
00025 extern "C" {
00026 #endif
00027
00028 /* Private includes -----*/
00029 /* USER CODE BEGIN Includes */
00030
00031 /* USER CODE END Includes */
00032
00033 /* Exported types -----*/
00034 /* USER CODE BEGIN ET */
00035
00036 /* USER CODE END ET */
00037
00038 /* Exported constants -----*/
00039 /* USER CODE BEGIN EC */
00040
00041 /* USER CODE END EC */
00042
00043 /* Exported macro -----*/
00044 /* USER CODE BEGIN EM */
00045
00046 /* USER CODE END EM */
00047
00048 /* Exported functions prototypes -----*/
00049 void NMI_Handler(void);
00050 void HardFault_Handler(void);
00051 void MemManage_Handler(void);
00052 void BusFault_Handler(void);
00053 void UsageFault_Handler(void);
00054 void SVC_Handler(void);
00055 void DebugMon_Handler(void);
00056 void PendSV_Handler(void);
00057 void SysTick_Handler(void);
00058
00059 void DMA1_Channel2_IRQHandler(void);
00060 void EXTI9_5_IRQHandler(void);
00061 void TIM1_UP_TIM16_IRQHandler(void);
00062
00063 void DMA1_Channel1_IRQHandler(void);
00064 void TIM1_TRG_COM_TIM17_IRQHandler(void);
00065
00066 void USART2_IRQHandler(void);
00067 void EXTI15_10_IRQHandler(void);
00068 void TIM6_DAC_IRQHandler(void);
00069 void TIM7_DAC_IRQHandler(void);
00070 /* USER CODE BEGIN EFP */
00071
00072 /* USER CODE END EFP */
00073
00074 #ifdef __cplusplus
00075 }
00076 #endif
00077
00078 #endif /* __STM32G4xx_IT_H */

```

6.18 Inc/tim.h File Reference

This file contains all the function prototypes for the [tim.c](#) file.

```
#include "main.h"
```

Functions

- void **MX_TIM1_Init** (void)
- void **MX_TIM3_Init** (void)
- void **MX_TIM7_Init** (void)
- void **MX_TIM16_Init** (void)
- void **MX_TIM17_Init** (void)
- void **HAL_TIM_MspPostInit** (TIM_HandleTypeDef *htim)

Variables

- TIM_HandleTypeDef **htim1**
- TIM_HandleTypeDef **htim3**
- TIM_HandleTypeDef **htim7**
- TIM_HandleTypeDef **htim16**
- TIM_HandleTypeDef **htim17**

6.18.1 Detailed Description

This file contains all the function prototypes for the [tim.c](#) file.

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6.18.2 Function Documentation

6.18.2.1 HAL_TIM_MspPostInit()

```
void HAL_TIM_MspPostInit (
    TIM_HandleTypeDef * htim)
```

TIM1 GPIO Configuration PB13 -----> TIM1_CH1N PB14 -----> TIM1_CH2N PB15 -----> TIM1_CH3N PA8 -----> TIM1_CH1 PA9 -----> TIM1_CH2 PA10 -----> TIM1_CH3

6.19 tim.h

[Go to the documentation of this file.](#)

```

00001 /* USER CODE BEGIN Header */
00019 /* USER CODE END Header */
00020 /* Define to prevent recursive inclusion -----*/
00021 #ifndef __TIM_H__
00022 #define __TIM_H__
00023
00024 #ifdef __cplusplus
00025 extern "C" {
00026 #endif
00027
00028 /* Includes -----*/
00029 #include "main.h"
00030
00031 /* USER CODE BEGIN Includes */
00032
00033 /* USER CODE END Includes */
00034
00035 extern TIM_HandleTypeDef htim1;
00036
00037 extern TIM_HandleTypeDef htim3;
00038
00039 extern TIM_HandleTypeDef htim7;
00040
00041 extern TIM_HandleTypeDef htim16;
00042
00043 extern TIM_HandleTypeDef htim17;
00044
00045 /* USER CODE BEGIN Private defines */
00046
00047 /* USER CODE END Private defines */
00048
00049 void MX_TIM1_Init(void);
00050 void MX_TIM3_Init(void);
00051 void MX_TIM7_Init(void);
00052 void MX_TIM16_Init(void);
00053 void MX_TIM17_Init(void);
00054 void HAL_TIM_MspPostInit(TIM_HandleTypeDef *htim);
00055
00056 /* USER CODE BEGIN Prototypes */
00057
00058 /* USER CODE END Prototypes */
00059
00060 #ifdef __cplusplus
00061 }
00062 #endif
00063
00064 #endif /* __TIM_H__ */
00065

```

6.20 Inc/usart.h File Reference

This file contains all the function prototypes for the [usart.c](#) file.

```
#include "main.h"
```

Functions

- void **MX_USART2_UART_Init** (void)
- void **MX_USART3_UART_Init** (void)

Variables

- UART_HandleTypeDef **huart2**
- UART_HandleTypeDef **huart3**

6.20.1 Detailed Description

This file contains all the function prototypes for the [usart.c](#) file.

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6.21 usart.h

[Go to the documentation of this file.](#)

```

00001 /* USER CODE BEGIN Header */
00019 /* USER CODE END Header */
00020 /* Define to prevent recursive inclusion -----*/
00021 #ifndef __USART_H__
00022 #define __USART_H__
00023
00024 #ifdef __cplusplus
00025 extern "C" {
00026 #endif
00027
00028 /* Includes -----*/
00029 #include "main.h"
00030
00031 /* USER CODE BEGIN Includes */
00032
00033 /* USER CODE END Includes */
00034
00035 extern UART_HandleTypeDef huart2;
00036
00037 extern UART_HandleTypeDef huart3;
00038
00039 /* USER CODE BEGIN Private defines */
00040
00041 /* USER CODE END Private defines */
00042
00043 void MX_USART2_UART_Init(void);
00044 void MX_USART3_UART_Init(void);
00045
00046 /* USER CODE BEGIN Prototypes */
00047
00048 /* USER CODE END Prototypes */
00049
00050 #ifdef __cplusplus
00051 }
00052 #endif
00053
00054 #endif /* __USART_H__ */
00055
```

6.22 button.h

```

00001 /*
00002  * button.h
00003  *
00004  * Created on: Nov 11, 2025
00005  * Author: nicolas
00006  */
00007
00008 #ifndef INC_BUTTON_H_
00009 #define INC_BUTTON_H_
00010
00011 #include "user_interface/shell.h"
00012
00013
00014 #endif /* INC_BUTTON_H_ */

```

6.23 led.h

```

00001 /*
00002  * led.h
00003  *
00004  * Created on: Nov 11, 2025
00005  * Author: nicolas
00006  */
00007
00008 #ifndef INC_LED_H_
00009 #define INC_LED_H_
00010
00011 #include "user_interface/shell.h"
00012
00013 #define TWO_ARGUMENTS 2
00014
00015 int led_init();
00016 int led_control(h_shell_t* h_shell, int argc, char** argv);
00017
00018 #endif /* INC_LED_H_ */

```

6.24 Inc/user_interface/shell.h File Reference

Header file for the shell module.

```

#include <stdint.h>
#include <stdio.h>
#include <string.h>
#include "main.h"
#include "stm32g4xx_hal.h"

```

Classes

- struct [shell_func_struct](#)
Structure representing a shell command.
- struct [drv_shell_struct](#)
Structure representing a shell driver.
- struct [h_shell_struct](#)
Structure representing a shell instance.

Macros

- #define [SHELL_ARGC_MAX](#) 8
- #define [SHELL_PRINT_BUFFER_SIZE](#) 64
- #define [SHELL_CMD_BUFFER_SIZE](#) 64
- #define [SHELL_FUNC_LIST_MAX_SIZE](#) 64
- #define **PROMPT** "MSC@SAC-TP:/"

Typedefs

- typedef int(* [shell_func_pointer_t](#)) (struct [h_shell_struct](#) *h_shell, int argc, char **argv)
Function pointer type for shell commands.
- typedef uint8_t(* [drv_shell_transmit_t](#)) (const char *pData, uint16_t size)
Function type for transmitting data.
- typedef uint8_t(* [drv_shell_receive_t](#)) (char *pData, uint16_t size)
Function type for receiving data.
- typedef struct [shell_func_struct](#) **shell_func_t**
Structure representing a shell command.
- typedef struct [drv_shell_struct](#) **drv_shell_t**
Structure representing a shell driver.
- typedef struct [h_shell_struct](#) **h_shell_t**
Structure representing a shell instance.

Functions

- void [shell_init](#) ([h_shell_t](#) *h_shell)
Initializes the shell instance.
- int [shell_add](#) ([h_shell_t](#) *h_shell, char *string_func_code, [shell_func_pointer_t](#) pfunc, char *description)
Adds a shell command to the instance.
- int [shell_run](#) ([h_shell_t](#) *h_shell)
Runs the shell.

Variables

- [h_shell_t](#) hshell1

6.24.1 Detailed Description

Header file for the shell module.

This file contains the declarations for the shell module, which provides a command-line interface.

6.24.2 Macro Definition Documentation

6.24.2.1 SHELL_ARGC_MAX

```
#define SHELL_ARGC_MAX 8
```

Maximum number of arguments for a shell command.

6.24.2.2 SHELL_CMD_BUFFER_SIZE

```
#define SHELL_CMD_BUFFER_SIZE 64
```

Size of the command buffer used for input.

6.24.2.3 SHELL_FUNC_LIST_MAX_SIZE

```
#define SHELL_FUNC_LIST_MAX_SIZE 64
```

Maximum number of registered shell functions.

6.24.2.4 SHELL_PRINT_BUFFER_SIZE

```
#define SHELL_PRINT_BUFFER_SIZE 64
```

Size of the print buffer used for output.

6.24.3 Typedef Documentation

6.24.3.1 drv_shell_receive_t

```
typedef uint8_t(* drv_shell_receive_t) (char *pData, uint16_t size)
```

Function type for receiving data.

This type represents a function pointer to a driver function for receiving data.

Parameters

<i>pData</i>	The pointer to the buffer to store the received data.
<i>size</i>	The maximum size of the buffer.

Returns

The number of bytes received.

6.24.3.2 drv_shell_transmit_t

```
typedef uint8_t(* drv_shell_transmit_t) (const char *pData, uint16_t size)
```

Function type for transmitting data.

This type represents a function pointer to a driver function for transmitting data.

Parameters

<i>pData</i>	The pointer to the data to be transmitted.
<i>size</i>	The size of the data to be transmitted.

Returns

The number of bytes transmitted.

6.24.3.3 shell_func_pointer_t

```
typedef int(* shell_func_pointer_t) (struct h_shell_struct *h_shell, int argc, char **argv)
```

Function pointer type for shell commands.

This type represents a function pointer to a shell command implementation.

Parameters

<i>h_shell</i>	The pointer to the shell instance.
<i>argc</i>	The number of command arguments.
<i>argv</i>	The array of command arguments.

Returns

The return value of the command execution.

6.24.4 Function Documentation

6.24.4.1 shell_add()

```
int shell_add (
    h_shell_t * h_shell,
    char * string_func_code,
    shell_func_pointer_t pfunc,
    char * description)
```

Adds a shell command to the instance.

This function adds a shell command to the shell instance.

Parameters

<i>h_shell</i>	The pointer to the shell instance.
<i>c</i>	The character trigger for the command.
<i>pfunc</i>	Pointer to the function implementing the command.
<i>description</i>	The description of the command.

Returns

0 on success, or a negative error code on failure.

6.24.4.2 shell_init()

```
void shell_init (
    h_shell_t * h_shell)
```

Initializes the shell instance.

This function initializes the shell instance by setting up the internal data structures and registering the help command.

Parameters

<i>h_shell</i>	The pointer to the shell instance.
----------------	------------------------------------

6.24.4.3 shell_run()

```
int shell_run (
    h_shell_t * h_shell)
```

Runs the shell.

This function runs the shell, processing user commands.

Parameters

<i>h_shell</i>	The pointer to the shell instance.
----------------	------------------------------------

Returns

Never returns, it's an infinite loop.

6.25 shell.h

[Go to the documentation of this file.](#)

```
00001
00002
00003 #ifndef SHELL_H_
00004 #define SHELL_H_
00005
00006 #include <stdint.h>
00007 #include <stdio.h>
00008 #include <string.h>
00009 #include "main.h"
00010 #include "stm32g4xx_hal.h"
00011
00012 #define SHELL_ARGC_MAX 8
00013 #define SHELL_PRINT_BUFFER_SIZE 64
00014 #define SHELL_CMD_BUFFER_SIZE 64
00015 #define SHELL_FUNC_LIST_MAX_SIZE 64
00016 #define PROMPT "MSC@SAC-TP:/"
00017
00018 struct h_shell_struct;
00019
00020 typedef int (*shell_func_pointer_t)(struct h_shell_struct* h_shell, int argc, char** argv);
00021
00022 typedef uint8_t (*drv_shell_transmit_t)(const char* pData, uint16_t size);
00023
00024 typedef uint8_t (*drv_shell_receive_t)(char* pData, uint16_t size);
```

```

00058
00062 typedef struct shell_func_struct
00063 {
00064     char* string_func_code;
00065     shell_func_pointer_t func;
00066     char* description;
00067 } shell_func_t;
00068
00072 typedef struct drv_shell_struct
00073 {
00074     drv_shell_transmit_t transmit;
00075     drv_shell_receive_t receive;
00076 } drv_shell_t;
00077
00081 typedef struct h_shell_struct
00082 {
00083     int func_list_size;
00084     shell_func_t func_list[SHELL_FUNC_LIST_MAX_SIZE];
00085     char print_buffer[SHELL_PRINT_BUFFER_SIZE];
00086     char cmd_buffer[SHELL_CMD_BUFFER_SIZE];
00087     drv_shell_t drv;
00088 } h_shell_t;
00089
00090 extern h_shell_t hshell1;
00091
00099 void shell_init(h_shell_t* h_shell);
00100
00112 int shell_add(h_shell_t* h_shell, char* string_func_code, shell_func_pointer_t pfunc, char*
    description);
00113
00122 int shell_run(h_shell_t* h_shell);
00123
00124 #endif /* SHELL_H_ */

```

6.26 Src/adc.c File Reference

This file provides code for the configuration of the ADC instances.

```
#include "adc.h"
```

Functions

- void [MX_ADC1_Init](#) (void)
- void [MX_ADC2_Init](#) (void)
- void [HAL_ADC_MspInit](#) (ADC_HandleTypeDef *adcHandle)
- void [HAL_ADC_MspDeInit](#) (ADC_HandleTypeDef *adcHandle)

Variables

- ADC_HandleTypeDef **hadc1**
- ADC_HandleTypeDef **hadc2**
- DMA_HandleTypeDef **hdma_adc1**

6.26.1 Detailed Description

This file provides code for the configuration of the ADC instances.

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6.26.2 Function Documentation

6.26.2.1 HAL_ADC_MspDeInit()

```
void HAL_ADC_MspDeInit (
    ADC_HandleTypeDef * adcHandle)
```

ADC1 GPIO Configuration PC2 -----> ADC1_IN8 PA1 -----> ADC1_IN2 PB0 -----> ADC1_IN15 PB1 -----> ADC1_IN12

ADC2 GPIO Configuration PC0 -----> ADC2_IN6 PC1 -----> ADC2_IN7 PC3 -----> ADC2_IN9 PA0 -----> ADC2_IN1

6.26.2.2 HAL_ADC_MspInit()

```
void HAL_ADC_MspInit (
    ADC_HandleTypeDef * adcHandle)
```

Initializes the peripherals clocks

ADC1 GPIO Configuration PC2 -----> ADC1_IN8 PA1 -----> ADC1_IN2 PB0 -----> ADC1_IN15 PB1 -----> ADC1_IN12

Initializes the peripherals clocks

ADC2 GPIO Configuration PC0 -----> ADC2_IN6 PC1 -----> ADC2_IN7 PC3 -----> ADC2_IN9 PA0 -----> ADC2_IN1

6.26.2.3 MX_ADC1_Init()

```
void MX_ADC1_Init (
    void )
```

Common config

Configure the ADC multi-mode

Configure Regular Channel

6.26.2.4 MX_ADC2_Init()

```
void MX_ADC2_Init (
    void )
```

Common config

Configure Regular Channel

6.27 Src/dma.c File Reference

This file provides code for the configuration of all the requested memory to memory DMA transfers.

```
#include "dma.h"
```

Functions

- void [MX_DMA_Init](#) (void)

6.27.1 Detailed Description

This file provides code for the configuration of all the requested memory to memory DMA transfers.

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6.27.2 Function Documentation

6.27.2.1 MX_DMA_Init()

```
void MX_DMA_Init (  
    void )
```

Enable DMA controller clock

6.28 Src/gpio.c File Reference

This file provides code for the configuration of all used GPIO pins.

```
#include "gpio.h"
```

Functions

- void [MX_GPIO_Init](#) (void)

6.28.1 Detailed Description

This file provides code for the configuration of all used GPIO pins.

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6.28.2 Function Documentation

6.28.2.1 MX_GPIO_Init()

```
void MX_GPIO_Init (  
    void )
```

Configure pins as Analog Input Output EVENT_OUT EXTI

6.29 Src/main.c File Reference

: Main program body

```
#include "main.h"  
#include "adc.h"  
#include "dma.h"  
#include "tim.h"  
#include "usart.h"  
#include "gpio.h"  
#include "app.h"  
#include "user_interface/shell.h"  
#include "acquisition/input_encoder.h"
```

Functions

- void [SystemClock_Config](#) (void)
System Clock Configuration.
- int [main](#) (void)
The application entry point.
- void [HAL_TIM_PeriodElapsedCallback](#) (TIM_HandleTypeDef *htim)
Period elapsed callback in non blocking mode.
- void [Error_Handler](#) (void)
This function is executed in case of error occurrence.

6.29.1 Detailed Description

: Main program body

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6.29.2 Function Documentation

6.29.2.1 Error_Handler()

```
void Error_Handler (  
    void )
```

This function is executed in case of error occurrence.

Return values

None	
------	--

6.29.2.2 HAL_TIM_PeriodElapsedCallback()

```
void HAL_TIM_PeriodElapsedCallback (  
    TIM_HandleTypeDef * htim)
```

Period elapsed callback in non blocking mode.

Note

This function is called when TIM6 interrupt took place, inside HAL_TIM_IRQHandler(). It makes a direct call to HAL_IncTick() to increment a global variable "uwTick" used as application time base.

Parameters

<i>htim</i>	: TIM handle
-------------	--------------

Return values

None	
------	--

6.29.2.3 main()

```
int main (
    void )
```

The application entry point.

Return values

<i>int</i>	
------------	--

6.29.2.4 SystemClock_Config()

```
void SystemClock_Config (
    void )
```

System Clock Configuration.

Return values

<i>None</i>	
-------------	--

Configure the main internal regulator output voltage

Initializes the RCC Oscillators according to the specified parameters in the RCC_OscInitTypeDef structure.

Initializes the CPU, AHB and APB buses clocks

6.30 Src/stm32g4xx_hal_msp.c File Reference

This file provides code for the MSP Initialization and de-Initialization codes.

```
#include "main.h"
```

Functions

- void [HAL_MspInit](#) (void)

6.30.1 Detailed Description

This file provides code for the MSP Initialization and de-Initialization codes.

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6.30.2 Function Documentation

6.30.2.1 HAL_MspInit()

```
void HAL_MspInit (  
    void )
```

Initializes the Global MSP. Disable the internal Pull-Up in Dead Battery pins of UCPD peripheral

6.31 Src/stm32g4xx_hal_timebase_tim.c File Reference

HAL time base based on the hardware TIM.

```
#include "stm32g4xx_hal.h"  
#include "stm32g4xx_hal_tim.h"
```

Functions

- HAL_StatusTypeDef [HAL_InitTick](#) (uint32_t TickPriority)
This function configures the TIM6 as a time base source. The time source is configured to have 1ms time base with a dedicated Tick interrupt priority.
- void [HAL_SuspendTick](#) (void)
Suspend Tick increment.
- void [HAL_ResumeTick](#) (void)
Resume Tick increment.

Variables

- TIM_HandleTypeDef **htim6**

6.31.1 Detailed Description

HAL time base based on the hardware TIM.

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6.31.2 Function Documentation

6.31.2.1 HAL_InitTick()

```
HAL_StatusTypeDef HAL_InitTick (  
    uint32_t TickPriority)
```

This function configures the TIM6 as a time base source. The time source is configured to have 1ms time base with a dedicated Tick interrupt priority.

Note

This function is called automatically at the beginning of program after reset by HAL_Init() or at any time when clock is configured, by HAL_RCC_ClockConfig().

Parameters

<i>TickPriority</i>	Tick interrupt priority.
---------------------	--------------------------

Return values

<i>HAL</i>	status
------------	--------

6.31.2.2 HAL_ResumeTick()

```
void HAL_ResumeTick (  
    void )
```

Resume Tick increment.

Note

Enable the tick increment by Enabling TIM6 update interrupt.

Parameters

<i>None</i>	
-------------	--

Return values

<i>None</i>	
-------------	--

6.31.2.3 HAL_SuspendTick()

```
void HAL_SuspendTick (
    void )
```

Suspend Tick increment.

Note

Disable the tick increment by disabling TIM6 update interrupt.

Parameters

None	
------	--

Return values

None	
------	--

6.32 Src/stm32g4xx_it.c File Reference

Interrupt Service Routines.

```
#include "main.h"
#include "stm32g4xx_it.h"
```

Functions

- void **NMI_Handler** (void)
This function handles Non maskable interrupt.
- void **HardFault_Handler** (void)
This function handles Hard fault interrupt.
- void **MemManage_Handler** (void)
This function handles Memory management fault.
- void **BusFault_Handler** (void)
This function handles Prefetch fault, memory access fault.
- void **UsageFault_Handler** (void)
This function handles Undefined instruction or illegal state.
- void **SVC_Handler** (void)
This function handles System service call via SWI instruction.
- void **DebugMon_Handler** (void)
This function handles Debug monitor.
- void **PendSV_Handler** (void)
This function handles Pendable request for system service.
- void **SysTick_Handler** (void)
This function handles System tick timer.
- void **DMA1_Channel2_IRQHandler** (void)
This function handles DMA1 channel2 global interrupt.

- void **EXTI9_5_IRQHandler** (void)
This function handles EXTI line[9:5] interrupts.
- void **TIM1_UP_TIM16_IRQHandler** (void)
This function handles TIM1 update interrupt and TIM16 global interrupt.
- void **TIM1_TRG_COM_TIM17_IRQHandler** (void)
This function handles TIM1 trigger and commutation interrupts and TIM17 global interrupt.
- void **USART2_IRQHandler** (void)
This function handles USART2 global interrupt / USART2 wake-up interrupt through EXTI line 26.
- void **EXTI15_10_IRQHandler** (void)
This function handles EXTI line[15:10] interrupts.
- void **TIM6_DAC_IRQHandler** (void)
This function handles TIM6 global interrupt, DAC1 and DAC3 channel underrun error interrupts.
- void **TIM7_DAC_IRQHandler** (void)
This function handles TIM7 global interrupt, DAC2 and DAC4 channel underrun error interrupts.

Variables

- DMA_HandleTypeDef **hdma_adc1**
- TIM_HandleTypeDef **htim1**
- TIM_HandleTypeDef **htim7**
- TIM_HandleTypeDef **htim17**
- UART_HandleTypeDef **huart2**
- TIM_HandleTypeDef **htim6**

6.32.1 Detailed Description

Interrupt Service Routines.

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6.33 Src/syscalls.c File Reference

STM32CubeIDE Minimal System calls file.

```
#include <sys/stat.h>
#include <stdlib.h>
#include <errno.h>
#include <stdio.h>
#include <signal.h>
#include <time.h>
#include <sys/time.h>
#include <sys/times.h>
```

Functions

- `int __io_putchar (int ch) __attribute__((weak))`
- `int __io_getchar (void)`
- `void initialise_monitor_handles ()`
- `int _getpid (void)`
- `int _kill (int pid, int sig)`
- `void _exit (int status)`
- `__attribute__((weak))`
- `int _close (int file)`
- `int _fstat (int file, struct stat *st)`
- `int _isatty (int file)`
- `int _lseek (int file, int ptr, int dir)`
- `int _open (char *path, int flags,...)`
- `int _wait (int *status)`
- `int _unlink (char *name)`
- `int _times (struct tms *buf)`
- `int _stat (char *file, struct stat *st)`
- `int _link (char *old, char *new)`
- `int _fork (void)`
- `int _execve (char *name, char **argv, char **env)`

Variables

- `char ** environ = __env`

6.33.1 Detailed Description

STM32CubeIDE Minimal System calls file.

Author

Auto-generated by STM32CubeIDE

```
For more information about which c-functions
need which of these lowlevel functions
please consult the Newlib libc-manual
```

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6.34 Src/sysmem.c File Reference

STM32CubeIDE System Memory calls file.

```
#include <errno.h>
#include <stdint.h>
```

Functions

- void * [_sbrk](#) (ptrdiff_t incr)
[_sbrk\(\)](#) allocates memory to the newlib heap and is used by malloc and others from the C library

6.34.1 Detailed Description

STM32CubeIDE System Memory calls file.

Author

Generated by STM32CubeIDE

For more information about which C functions
need which of these lowlevel functions
please consult the newlib libc manual

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6.34.2 Function Documentation

6.34.2.1 [_sbrk\(\)](#)

```
void * _sbrk (
    ptrdiff_t incr)
```

[_sbrk\(\)](#) allocates memory to the newlib heap and is used by malloc and others from the C library

```
* #####
* # .data # .bss #          newlib heap          #          MSP stack          #
* #          #          #          # Reserved by _Min_Stack_Size #
* #####
* ^-- RAM start          ^-- _end          _estack, RAM end --^
*
```

This implementation starts allocating at the '_end' linker symbol The '_Min_Stack_Size' linker symbol reserves a memory for the MSP stack The implementation considers '_estack' linker symbol to be RAM end NOTE: If the MSP stack, at any point during execution, grows larger than the reserved size, please increase the '_Min_Stack_Size'.

Parameters

<i>incr</i>	Memory size
-------------	-------------

Returns

Pointer to allocated memory

6.35 Src/system_stm32g4xx.c File Reference

CMSIS Cortex-M4 Device Peripheral Access Layer System Source File.

```
#include "stm32g4xx.h"
```

Macros

- #define [HSE_VALUE](#) 24000000U
- #define [HSI_VALUE](#) 16000000U

Functions

- void [SystemInit](#) (void)
Setup the microcontroller system.
- void [SystemCoreClockUpdate](#) (void)
Update SystemCoreClock variable according to Clock Register Values. The SystemCoreClock variable contains the core clock (HCLK), it can be used by the user application to setup the SysTick timer or configure other parameters.

Variables

- uint32_t **SystemCoreClock** = [HSI_VALUE](#)
- const uint8_t **AHBPrescTable** [16] = {0U, 0U, 0U, 0U, 0U, 0U, 0U, 0U, 1U, 2U, 3U, 4U, 6U, 7U, 8U, 9U}
- const uint8_t **APBPrescTable** [8] = {0U, 0U, 0U, 0U, 1U, 2U, 3U, 4U}

6.35.1 Detailed Description

CMSIS Cortex-M4 Device Peripheral Access Layer System Source File.

Author

MCD Application Team

This file provides two functions and one global variable to be called from user application:

- [SystemInit\(\)](#): This function is called at startup just after reset and before branch to main program. This call is made inside the "startup_stm32g4xx.s" file.
- SystemCoreClock variable: Contains the core clock (HCLK), it can be used by the user application to setup the SysTick timer or configure other parameters.
- [SystemCoreClockUpdate\(\)](#): Updates the variable SystemCoreClock and must be called whenever the core clock is changed during program execution.

After each device reset the HSI (16 MHz) is used as system clock source. Then [SystemInit\(\)](#) function is called, in "startup_stm32g4xx.s" file, to configure the system clock before to branch to main program.

6.35.2 This file configures the system clock as follows:**6.35.2.1 System Clock source | HSI****6.35.2.2 SYSCCLK(Hz) | 16000000****6.35.2.3 HCLK(Hz) | 16000000****6.35.2.4 AHB Prescaler | 1****6.35.2.5 APB1 Prescaler | 1****6.35.2.6 APB2 Prescaler | 1****6.35.2.7 PLL_M | 1****6.35.2.8 PLL_N | 16****6.35.2.9 PLL_P | 7****6.35.2.10 PLL_Q | 2****6.35.2.11 PLL_R | 2****6.35.2.12 Require 48MHz for RNG | Disabled**

=====

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6.36 Src/tim.c File Reference

This file provides code for the configuration of the TIM instances.

```
#include "tim.h"
```

Functions

- void **MX_TIM1_Init** (void)
- void **MX_TIM3_Init** (void)
- void **MX_TIM7_Init** (void)
- void **MX_TIM16_Init** (void)
- void **MX_TIM17_Init** (void)
- void **HAL_TIM_PWM_MspInit** (TIM_HandleTypeDef *tim_pwmHandle)
- void **HAL_TIM_Encoder_MspInit** (TIM_HandleTypeDef *tim_encoderHandle)
- void **HAL_TIM_Base_MspInit** (TIM_HandleTypeDef *tim_baseHandle)
- void **HAL_TIM_MspPostInit** (TIM_HandleTypeDef *timHandle)
- void **HAL_TIM_PWM_MspDeInit** (TIM_HandleTypeDef *tim_pwmHandle)
- void **HAL_TIM_Encoder_MspDeInit** (TIM_HandleTypeDef *tim_encoderHandle)
- void **HAL_TIM_Base_MspDeInit** (TIM_HandleTypeDef *tim_baseHandle)

Variables

- TIM_HandleTypeDef **htim1**
- TIM_HandleTypeDef **htim3**
- TIM_HandleTypeDef **htim7**
- TIM_HandleTypeDef **htim16**
- TIM_HandleTypeDef **htim17**

6.36.1 Detailed Description

This file provides code for the configuration of the TIM instances.

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6.36.2 Function Documentation

6.36.2.1 HAL_TIM_Base_MspDeInit()

```
void HAL_TIM_Base_MspDeInit (
    TIM_HandleTypeDef * tim_baseHandle)
```

Uncomment the line below to disable the "TIM1_TRG_COM_TIM17_IRQn" interrupt Be aware, disabling shared interrupt may affect other IPs

6.36.2.2 HAL_TIM_Encoder_MspDeInit()

```
void HAL_TIM_Encoder_MspDeInit (
    TIM_HandleTypeDef * tim_encoderHandle)
```

TIM3 GPIO Configuration PA4 -----> TIM3_CH2 PA6 -----> TIM3_CH1

6.36.2.3 HAL_TIM_Encoder_MspInit()

```
void HAL_TIM_Encoder_MspInit (
    TIM_HandleTypeDef * tim_encoderHandle)
```

TIM3 GPIO Configuration PA4 -----> TIM3_CH2 PA6 -----> TIM3_CH1

6.36.2.4 HAL_TIM_MspPostInit()

```
void HAL_TIM_MspPostInit (
    TIM_HandleTypeDef * timHandle)
```

TIM1 GPIO Configuration PB13 -----> TIM1_CH1N PB14 -----> TIM1_CH2N PB15 -----> TIM1_CH3N PA8 -----> TIM1_CH1 PA9 -----> TIM1_CH2 PA10 -----> TIM1_CH3

6.36.2.5 HAL_TIM_PWM_MspDeInit()

```
void HAL_TIM_PWM_MspDeInit (
    TIM_HandleTypeDef * tim_pwmHandle)
```

Uncomment the line below to disable the "TIM1_TRG_COM_TIM17_IRQn" interrupt Be aware, disabling shared interrupt may affect other IPs

6.37 Src/usart.c File Reference

This file provides code for the configuration of the USART instances.

```
#include "usart.h"
```

Functions

- void **MX_USART2_UART_Init** (void)
- void **MX_USART3_UART_Init** (void)
- void **HAL_UART_MspInit** (UART_HandleTypeDef *uartHandle)
- void **HAL_UART_MspDeInit** (UART_HandleTypeDef *uartHandle)

Variables

- UART_HandleTypeDef **huart2**
- UART_HandleTypeDef **huart3**

6.37.1 Detailed Description

This file provides code for the configuration of the USART instances.

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6.37.2 Function Documentation**6.37.2.1 HAL_UART_MspDeInit()**

```
void HAL_UART_MspDeInit (
    UART_HandleTypeDef * uartHandle)
USART2 GPIO Configuration PA2 -----> USART2_TX PA3 -----> USART2_RX
USART3 GPIO Configuration PC10 -----> USART3_TX PC11 -----> USART3_RX
```

6.37.2.2 HAL_UART_MspInit()

```
void HAL_UART_MspInit (
    UART_HandleTypeDef * uartHandle)
Initializes the peripherals clocks
USART2 GPIO Configuration PA2 -----> USART2_TX PA3 -----> USART2_RX
Initializes the peripherals clocks
USART3 GPIO Configuration PC10 -----> USART3_TX PC11 -----> USART3_RX
```

6.38 Src/user_interface/shell.c File Reference

Implementation file for the shell module.

```
#include "user_interface/shell.h"
```

Functions

- void [shell_init](#) ([h_shell_t](#) *h_shell)
Initializes the shell instance.
- int [shell_add](#) ([h_shell_t](#) *h_shell, char *string_func_code, [shell_func_pointer_t](#) pfunc, char *description)
Adds a shell command to the instance.
- int [shell_run](#) ([h_shell_t](#) *h_shell)
Runs the shell.

Variables

- [h_shell_t](#) **hshell1**

6.38.1 Detailed Description

Implementation file for the shell module.

This file contains the implementation of the shell module, which provides a command-line interface.

6.38.2 Function Documentation

6.38.2.1 shell_add()

```
int shell_add (  
    h_shell_t * h_shell,  
    char * string_func_code,  
    shell_func_pointer_t pfunc,  
    char * description)
```

Adds a shell command to the instance.

This function adds a shell command to the shell instance.

Parameters

<i>h_shell</i>	The pointer to the shell instance.
<i>c</i>	The character trigger for the command.
<i>pfunc</i>	Pointer to the function implementing the command.
<i>description</i>	The description of the command.

Returns

0 on success, or a negative error code on failure.

6.38.2.2 shell_init()

```
void shell_init (  
    h_shell_t * h_shell)
```

Initializes the shell instance.

This function initializes the shell instance by setting up the internal data structures and registering the help command.

Parameters

<i>h_shell</i>	The pointer to the shell instance.
----------------	------------------------------------

6.38.2.3 shell_run()

```
int shell_run (  
    h_shell_t * h_shell)
```

Runs the shell.

This function runs the shell, processing user commands.

Parameters

<i>h_shell</i>	The pointer to the shell instance.
----------------	------------------------------------

Returns

Never returns, it's an infinite loop.

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